CADFEM-Bench v1.2: published-source CAD + FEM pilot

2026-10-14. **114 active tasks** with **294 graded outputs** in 9 engineering families; 54 tasks held with reasons; 190 locally computed v1.0 targets excluded. Every graded target is a value published by its source (saved solver results, publisher tables and evaluators, result legends, NAFEMS targets); none was computed for this release. This is a **source-backed pilot**, not a certified benchmark: 12 tasks are independently reproduced, most tolerances are provisional, and 84 of 114 tasks come from three parametric families (SimJEB, AgentFEM, GeoPINONet). Frictional contact, plasticity, forming, crash, fatigue and coupled assembly mechanics are not covered by the active tasks.

Tracks. A: the model receives the CAD and the source set-up and must compute the requested results (117 tasks). B: the model receives the dimensioned drawing, builds the CAD and analyses it (1 task). **Tiers.** source_backed_pilot (numerical published reference), figure_reference (legend value, set-up partly from the source's BC figure), control (NAFEMS).

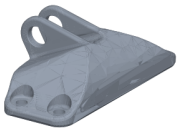
engineering family	tracks	tiers	tasks	graded outputs	distinct CAD	physics	reference class
simjeb-bracket	A	source_backed_pilot	36	144	36	linear static, multi-load-case	published_table
agentfem-bracket	A	source_backed_pilot	32	64	32	linear static	published_table
geopinonet-holed-plate	A	source_backed_pilot	16	32	16	linear static	published_raw_field
cycloidal-drive-components	A	source_backed_pilot	11	11	11	linear static	published_raw_field
morphing-wing-plates	A	source_backed_pilot	8	24	8	modal	published_table
rigging-plate	A/B	figure_reference	6	6	1	linear static	published_legend
nafems-calibration	A	control	3	3	4	linear static, plane stress, thermoelastic	benchmark_target
cadworld-freecad-fem	A	source_backed_pilot	1	4	1	linear static	publisher_evaluator
feenox-valve-transient	A	source_backed_pilot	1	6	1	multi-material, nonlinear material, transient thermal	published_table
total			114	294	107		

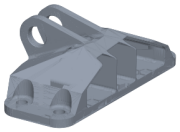
Changes in v1.2 (2026-10-14). The published cycloidal displacements are coarse-mesh values (the coupler converges about 5 % above its reference, outside the old 5 % tolerance). An independent FEniCSx convergence study now sets each part's tolerance to the observed published-vs-converged difference plus twice the last refinement change (at least 5 %); 11 parts are admitted this way and 4 held. Published values are unchanged. AgentFEM fresh_test_000 is recorded as reproduced (follow-up study).

Changes from v1.0 (audit of 2026-10-13). (1) Scoring separates *numerical_passed* from *workflow_verified*; a pass needs both, and workflow verification needs eight artifacts (results, script, STEP, mesh, fields, solver log, refinement study, environment) plus a trusted review bound to task, run and artifact hashes. (2) The 190 targets computed by the local SimJEB grader are removed; SimJEB analysis keeps SimJEB's published displacement maxima and the 40 design tasks are held. (3) The two SimJEB STEP files that fail OpenCASCADE validity are held. (4) The rigging drawing task has a volume-overlap gate against the original CAD after alignment over the 24 axis rotations; the other 8 reconstruction tasks are held. (5) CADWorld's 'Tresca' column is (s1 - s3)/2 and is now requested as maximum shear stress. (6) Malformed submissions are rejected per task without stopping the run. (7) Tiers are renamed to what they show, and the original source files for every reference ship in an evaluator-only evidence bundle with hashes.

Verification at build time. 148 input files and 382 evidence files hash-checked; all 107 distinct CAD files valid in OpenCASCADE; no reference number in any prompt or text input; no verifier target. Scorer probes: exact answers alone give 114 numerical passes and 0 passes; answers scaled by 1.3 give 0 numerical passes; four malformed submissions are rejected without stopping the run; a reviewed probe passes and is rejected once an artifact changes; the rigging gate gives symmetric difference 0 for the source CAD rotated and moved, and 0.246 for it scaled by 3 % (limit 0.02). Earlier set-up checks still hold: cycloidal fixed node sets lie on the stated faces and per-face loads sum to the deck resultants; GeoPINONet section forces are -1000 N; AgentFEM CAD scaled to mm matches the FEM nodes; the valve's Gmsh surface tags equal the OpenCASCADE face indices.

GE jet-engine bracket: four-case analysis of real competition designs





Ti-6Al-4V bracket, four GE load cases at the clevis pin (RBE3), rigid fully fixed bolts. Output per case: maximum displacement magnitude, graded against SimJEB's published OptiStruct maxima (10 %). Peak stress and the pass verdict are no longer graded (SimJEB does not publish them). Held: analyses 15 and 120 (invalid STEP) and all 40 design/redesign/lighten tasks (graded by the local FEM grader).

Tasks: 36. **Licence:** Dataverse CC0 / README ODC-By; GE bracket CAD from GrabCAD (non-commercial). Local research use. **Source:** SimJEB: simulated jet engine bracket dataset (Whalen, Beyene, Mueller, CGF 2021) / GE bracket challenge.

Example A-SIMJEB-analysis-0 (hidden references; 1 evidence files):

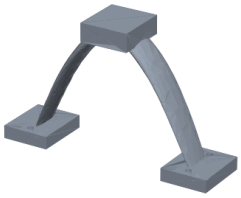
output	unit	reference	tolerance	class	evidence / provenance
vertical_max_displacement_mm	mm	0.321826	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_ver_magdisp (OptiStruct)
horizontal_max_displacement_mm	mm	0.239873	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_hor_magdisp (OptiStruct)
diagonal_max_displacement_mm	mm	0.165667	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_dia_magdisp (OptiStruct)
torsion_max_displacement_mm	mm	0.174951	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_tor_magdisp (OptiStruct)

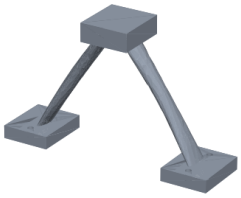
Example prompt (A-SIMJEB-analysis-0), first lines:

```
Design task: the GE jet engine bracket. Units: millimetres, newtons, MPa. Material: Ti-6Al-4V (E = 113.8 GPa, nu
= 0.342, density 4.43 g/cm^3, yield strength 903 MPa). The bracket is bolted to a flat mounting surface, the
plane z = 0, with nothing below it, and carries a clevis for a pin.
Interfaces: (a) four vertical (+z) bolt holes of diameter 10.3124 mm (0.406 in) at (x, y) = (0.0000, 0.0000);
(52.0472, 1.5805); (0.0000, -148.1248); (38.0833, -146.9684); each hole wall must start within 1.0 mm of z = 0
and be at least 5.5 mm long; the bolts are rigid and fully fixed. (b) a clevis of two lugs, each with a bore of
diameter 19.1008 mm on the pin axis through (-20.9236, -74.6978, 44.7246) with direction (-0.030352, 0.999539,
0); measured along the axis from that point, the lug bores span 11.112 to 17.463 mm and -17.463 to -11.113 mm;
the gap between the lugs must stay empty.
Design envelope: x -41 to 72, y -168 to 22, z 0 to 65 mm. Minimum wall / feature thickness 1.27 mm (at most 2 % of
the surface may be thinner). One solid.
Loads, four separate linear static cases, each acting at the point (-21.0359, -75.0647, 44.8010) and spread ...
```

Flags: Original published displacement targets only; source-model reproduction and tolerance remain provisional.; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

AgentFEM double-arm support brackets (publisher test split)





Aluminium bracket (E 70 GPa), foot undersides fixed, uniform downward seat traction of 1000 or 10000 N. Outputs: mean seat deflection and maximum displacement (publisher summary, 10 %; publisher mesh not converged). The source STEP declared mm but held metre values; the shipped copy is scaled to true mm and checked against the FEM nodes.

Tasks: 32. **Licence:** CC BY 4.0. **Source:** AgentFEM Bracket Elasticity (Haoming Luo, 2026).

Example A-AGENTFEM-fresh_test_000 (hidden references; 14 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
seat_mean_displacement_mm	mm	0.0983548	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json
max_displacement_mm	mm	0.175628	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json

Example prompt (A-AGENTFEM-fresh_test_000), first lines:

```
Support bracket (one member of a parametric double-arm bracket family). geometry.step holds one solid in
millimetres (height 189 mm).

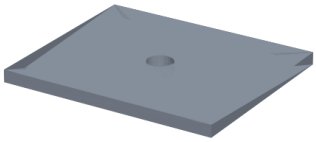
Material: linear isotropic elastic, E = 70 GPa, Poisson ratio 0.33 (density 2700 kg/m^3, gravity not applied).
Supports: the foot undersides, i.e. every face lying in the plane z = 0, are fully fixed (all displacement
components zero).
Load: a uniform downward traction of 1.98728 MPa in the -z direction over the top seat face (the face in the
plane z = 189 mm); total 10000 N.
Analysis: 3-D linear elasticity, small displacements. The publisher used quadratic tetrahedra with a nominal 9 mm
mesh.

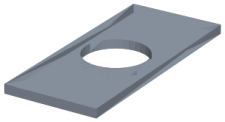
Outputs:
- seat_mean_displacement_mm [mm]: mean of the downward displacement -u_z over the top seat face (z = 189 mm)
- max_displacement_mm [mm]: maximum displacement magnitude anywhere in the bracket

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: source STEP unit error fixed: shipped copy scaled x1000 to true millimetres and checked against the FEM nodes; one parametric family: report as a family macro-average; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.; Independently reproduced (2026-10-14 follow-up, 5 meshes): max displacement within 0.006 %, seat mean within 0.47 % of the published values.

GeoPINONet holed steel plates (validation split)





Structural steel, one end fixed, 1000 N on the opposite end face: compression (-y) and lateral (+x). Outputs: maximum displacement per case from the published ANSYS nodal CSVs (5 %). The archive's fixed/load STL names are swapped relative to the fields; supports and loads are stated from the verified fields.

Tasks: 16. **Licence:** CC BY 4.0. **Source:** GeoPINONet holed-plate FEM dataset (Urrutia Valdes, La Fe-Perdomo, Rodriguez Soto).

Example A-GEOPINO-PLATE-1 (hidden references; 4 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_compression_mm	mm	0.000439557	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)
max_displacement_lateral_mm	mm	0.00303832	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)

Example prompt (A-GEOPINO-PLATE-1), first lines:

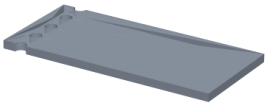
```
Holed steel plate. part.step is one solid in millimetres: a rectangular plate 176.914 mm (x) by 148.475 mm (y) by
10.000 mm (z) with a through hole along z.
Material: ANSYS default Structural Steel, linear isotropic elastic, E = 200 GPa, Poisson ratio 0.3. Linear
static, small displacements, no gravity.
Supports: the end face at y = -74.2373 mm is fully fixed (all displacement components zero).
Two separate load cases, each a total force of 1000 N applied as a uniform traction over the opposite end face (y
= 74.2373 mm):
(1) compression: 1000 N along -y (towards the fixed end);
(2) lateral bending: 1000 N along +x (in-plane shear of the free end).

Outputs:
- max_displacement_compression_mm [mm]: maximum displacement magnitude, load case 1
- max_displacement_lateral_mm [mm]: maximum displacement magnitude, load case 2

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: source errors not carried over: BCs stated from the verified fields (archive STL names are swapped), thickness from the CAD (README says 20 mm, CAD and FEM 10 mm); No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

Active-camber morphing-wing cantilever plates (modal)





FDM-printed PLA plates, uniform and with a 1 mm flexible strip, clamped at the end of a 19 mm grip. Outputs: first bending, first torsion and second bending frequencies from the publisher workbook (Abaqus thick shell / Nastran plate models, 5 %). The references are shell idealisations of solid CAD.

Tasks: 8. **Licence:** CC BY 4.0. **Source:** Experimental and numerical data of candidate structural models for active camber morphing wings (Huang & Zhang).

Example A-MORPHWING-Test1 (hidden references; 46 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
f_1B_Hz	Hz	109.05	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S4 (Abaqus thick shell S8R)
f_1T_Hz	Hz	416.67	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S5 (Abaqus thick shell S8R)
f_2B_Hz	Hz	672.94	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S6 (Abaqus thick shell S8R)

Example prompt (A-MORPHWING-Test1), first lines:

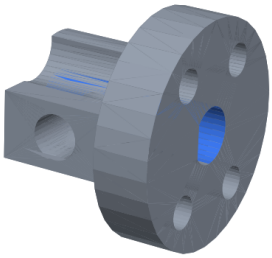
```
Cantilever plate specimen for an active-camber morphing-wing study (FDM-printed PLA): a uniform 4 mm plate.
part.step is one solid in millimetres, 109 x 50 x 4 mm, with a 19 mm clamping grip at x = 0-19 mm (the scalloped
clamp region).
Model the vibrating structure only: the free part x >= 19 mm, clamped (all displacements and rotations zero)
along its whole cross-section at x = 19 mm. The grip is held in the fixture and is not part of the model.
Material: PLA treated as isotropic linear elastic, E = 2107.5 MPa, Poisson ratio 0.3, density 1180 kg/m^3.
Analysis: undamped natural frequencies (modal analysis), no prestress, no gravity. Identify the modes by their
shapes.

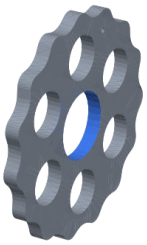
Outputs:
- f_1B_Hz [Hz]: first out-of-plane bending mode
- f_1T_Hz [Hz]: first torsion mode about the span (x) axis
- f_2B_Hz [Hz]: second out-of-plane bending mode

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: reference is a shell-model solution of the free length; solid models converge to it within about 1-2 %; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

3D-printable cycloidal gearbox: component static analyses





FreeCAD/CalculiX studies of 13 printed PETG-CF parts and 2 steel pins, each analysed alone (no gearbox contact). Blue: fixed faces; red: loaded faces (uniform traction, resultant from the source-corrected deck). Output: maximum displacement from the saved FRD files, with a tolerance set by an independent convergence study (see the study page); stress not graded (source error estimate 11-49 %).

Tasks: 11. **Licence:** No repository license; FCStd metadata 'All rights reserved'. Local research only. **Source:** 3D-printable cycloidal drive FEA (jagathsrujan/3d-printable-cycloidal-drive).

Example A-CYC-Coupler-PETGCF (hidden references; 15 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_mm	mm	0.00761747	rel 0.06	published_raw_field	fem/petg_cf/FEM_Coupler_PETGCF/Coupler_fixed.frd (saved CalculiX FRD of the source-corrected deck)

Example prompt (A-CYC-Coupler-PETGCF), first lines:

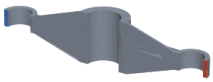
```
Component analysis: the output coupler of an open-source, 3D-printable 15:1 cycloidal gearbox. part.step holds
the single solid in millimetres (33 faces, volume 2063 mm^3). faces.json lists every face with its index, type,
area, centroid and normal/axis; use the geometry, not a CAD system's own face numbering.

Material: PETG-CF, linear isotropic elastic, E = 3000 MPa, Poisson ratio 0.36.
Analysis: linear static, small displacements, no gravity, no contact.
Supports: all three displacement components are zero on face 5: cylinder, area 33.52 mm^2, centroid (76.619,
-0.908, 22.737) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through (79.025, -0.0, 20.5) (hole/bore surface); face
6: cylinder, area 117.81 mm^2, centroid (76.525, 1.588, 20.397) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through
(79.025, -0.0, 20.5) (hole/bore surface).
Load: total force (100.052, 0.000, 0.000) N applied as a uniform surface traction (constant force per unit area)
over the loaded face(s): face 14 carries (100.052, 0.000, 0.000) N.
For reference, the source solved this with second-order tetrahedra (Netgen, maximum element size 2 mm).

Outputs:
- max_displacemen ...
```

Flags: source author corrected FreeCAD CLOAD units by 1000x; loads stated from the corrected deck; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.; Tolerance 6 % from an independent convergence study: finest independent 0.00790579 mm vs published 0.00761747 mm (3.8 %), last refinement change 0.87 %.

CADWorld FreeCAD FEM task (publisher mesh protocol)



Publisher geometry, material, constraints and mesh protocol (Gmsh, first-order tetrahedra); references are the values the publisher's evaluator checks (5 %). The publisher's 'Tresca' column is maximum shear stress (s1 - s3)/2 and is now requested under that name. The task reproduces the source's linear model (31 mm deflection of a 260 mm part, stresses above yield); it is not a strength assessment.

Tasks: 1. **Licence:** Apache-2.0 (publisher HF metadata). **Source:** CADWorld FEM benchmark (Zdong104/CADWORLD; HF Zihan1004/CADWORLD).

Example A-CADWORLD-001 (hidden references; 8 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_mm	mm	31.145	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Displacement Magnitude' ma
max_von_mises_MPa	MPa	705.453	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'von Mises Stress' max (pub
max_shear_stress_MPa	MPa	397.017	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Tresca Stress' max (publis
max_principal_MPa	MPa	863.519	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Major Principal Stress' ma

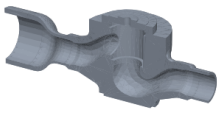
Example prompt (A-CADWORLD-001), first lines:

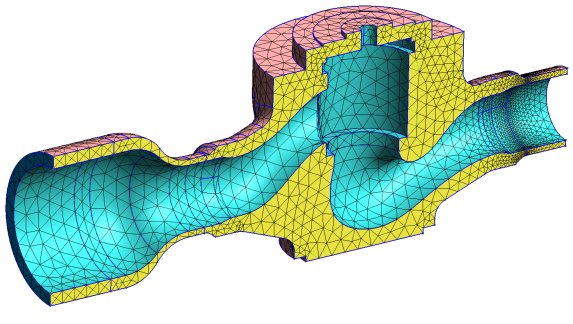
```
FreeCAD/CalculiX benchmark part (Multi-feature aluminium part, end load). part.step is the single solid in
millimetres (38 faces, bounding box [-130.0, -0.0, -30.0, 130.0, 30.0, 30.0] mm). faces.json describes every
face.

Material: AlMg3F24, linear isotropic elastic, E = 70000 MPa, Poisson ratio 0.3.
Analysis: linear static, small displacements, no gravity.
Supports: fixed (all displacement components zero) on face 6: plane, area 100.00 mm^2, centroid (-127.5, 0.0,
0.0) mm, outward normal (-0.0, -1.0, -0.0).
Load: total force 200 N along (0, 1, 0) applied uniformly over face 38: plane, area 100.00 mm^2, centroid (127.5,
-0.0, 0.0) mm, outward normal (-0.0, -1.0, -0.0).
Mesh protocol of the publisher's reference solution (results are compared under this protocol): Gmsh 3-D mesh,
first-order (linear) tetrahedra, maximum element size 2.0 mm, minimum 0.5 mm, CalculiX solver. Report nodal
extrema of that solution.
The publisher column named Tresca Stress is maximum shear stress, (sigma1 - sigma3)/2. Report this
half-difference as max_shear_stress_MPa, not the full principal-stress difference. This task reproduces the
source linear model ...
```

Flags: protocol-bound: publisher mesh (first-order tets) fixes the reference; publisher tolerance 5 %; linear-elastic stresses exceed material yield; computational benchmark, not a strength verdict; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

Power-plant valve: nonlinear transient heat conduction





Carbon-steel nozzle and stainless bodies (4 solids), ASME temperature-dependent properties, internal temperature ramp 40-250-40 C over 1000 s, weak external convection. Outputs: through-wall temperature differences at two stress classification lines (10 %) and outer-wall temperatures at 100 s, from the published FeenoX history.

Tasks: 1. **Licence:** GPL-3.0. **Source:** FeenoX tutorial #3: heat conduction (valve with carbon and stainless steel parts).

Example A-VALVE-transient (hidden references; 11 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
scl2_dT_max_K	K	14.029	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_dT_min_K	K	-15.183	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_max_K	K	26.128	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_min_K	K	-28.159	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_T_outer_100s_C	C	235.837	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_T_outer_100s_C	C	223.738	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)

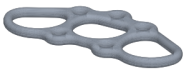
Example prompt (A-VALVE-transient), first lines:

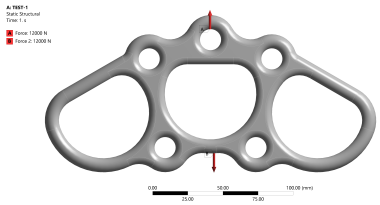
```
Nonlinear transient heat conduction in a power-plant valve: a carbon-steel nozzle welded to stainless-steel valve
bodies (half model, symmetry plane y = 0). valve.brep holds 4 solids in millimetres (112 faces). Solid 2 (the
nozzle, x from 0 to 65 mm) is carbon steel; solids 1, 3 and 4 are stainless steel SA-312. faces.json lists every
face with its index and its boundary group; valve-surfaces.png and valve-materials.png are the source's figures.

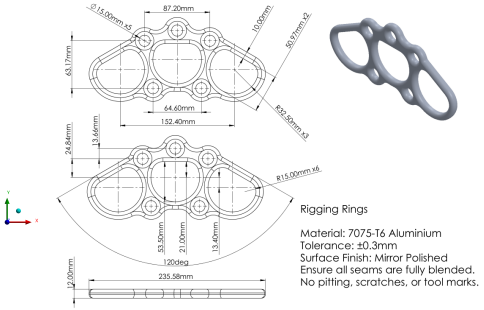
Materials: temperature-dependent conductivity k(T) and thermal diffusivity kappa(T) from asme-properties.fee
(ASME II Part D tables, T in deg C): carbon steel uses k_carbon and kappa_carbon, stainless uses k_312 and
kappa_312. As in the source, k from the tables is in W/(m K) and is multiplied by 1e-3 to W/(mm K); kappa is used
as tabulated (mm^2/s). The volumetric heat capacity follows as k/kappa.
Boundary conditions: 'internal' faces: prescribed temperature T = Tint(t) from internal-temperature.csv (linear
interpolation: 40 C at t = 0, 250 C from 100 s to 500 s, back to 40 C at 600 s, held to 1000 s). 'external' faces:
convection h = 1e-6 W/(mm^2 K) to 50 C. 'symmetry' and both end faces: ...
```

Flags: through-wall differences use the published 192-sample history; 10 % tolerance on differences; temperatures at 100 s graded on the rise above the 40 C initial state; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

7075-T6 rigging plate: five ANSYS load tests and the drawing-to-CAD task (figure_reference)







Supplied STEP (track A) or the dimensioned manufacturing drawing (track B) plus the source's boundary-condition figure per test. Output: maximum von Mises stress from the original legend (20 %). Track B: the built part must overlap the original CAD within 2 % symmetric difference after alignment. Material card and remote-coupling details are not published; tests 4 and 5 are left out (conflicting load labels; revised TEST-5R results).

Tasks: 6. **Licence:** CERN-OHL-S-2.0. **Source:** OpenRiggingPlate (mpangch).

Example A-RIGGING-test1 (hidden references; 5 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_von_mises_MPa	MPa	526.74	rel 0.2	published_legend	FEA Analysis/Test 1 (2697.7 lbf)/Stress.png legend '526.74 Max'

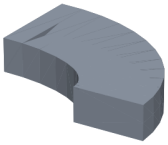
Example prompt (A-RIGGING-test1), first lines:

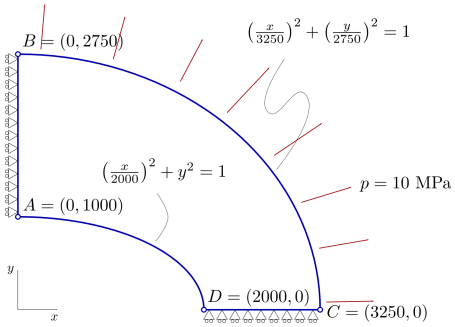
```
Hard-anodised 7075-T6 rigging plate, load test 1 of the source's seven ANSYS load cases (Test 1 (2697.7 lbf)).
riggingplate.step is the single solid in millimetres (235.58 x 111.90 x 12 mm, 145 faces).
Material: aluminium 7075-T6, linear isotropic elastic (standard handbook values; the source did not publish its
card, and the graded stress does not depend on Young's modulus for these loads).
Analysis: linear static, small displacements (ANSYS Mechanical in the source).
Boundary conditions: exactly as shown in boundary-conditions.png (the source's own ANSYS figure): each labelled
scope (A, B, ...) is a face or set of faces of the plate; 'Force' applies the stated total force uniformly
distributed over the scoped faces in the drawn direction; 'Remote Displacement' fixes the scoped faces through a
rigid remote point (all displacements and rotations zero). Where only two equal and opposite forces are applied
and no support is drawn, the plate is self-equilibrated; remove rigid-body motion without adding load (e.g. weak
springs or inertia relief), as the source did.
Legend of this test's figure: A Force: 12000 N; B; Force 2: 12000 N.

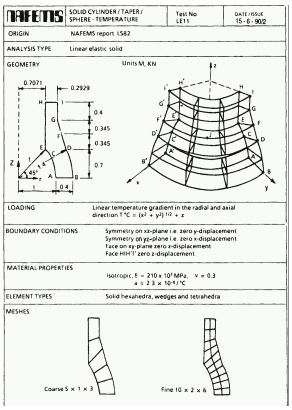
Ou ...
```

Flags: load scopes given by the source figure, not by face IDs; reference is a rounded legend maximum; peak at load-face edges is mesh-sensitive (20 % tolerance); No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

NAFEMS calibration (control)







NAFEMS LE1 (elliptic membrane), LE10 (thick elliptic plate) and LE11 (thermoelastic cylinder/taper/sphere) with supplied geometry. References are the FeenoX outputs, equal to the NAFEMS targets (5 %). Answers may be recalled; calibration only. The three drawing-to-CAD variants and the five FEMCAD-Hard tasks are held.

Tasks: 3. **Licence:** GPL-3.0-or-later (examples). **Source:** FeenoX NAFEMS mechanical examples (seamplex/feenox).

Example A-NAFEMS-LE10 (hidden references; 6 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
sigma_yy_at_D	MPa	-5.37968	rel 0.05	benchmark_target	Published terminal block in section NAFEMS LE10 thick elliptic plate

Example prompt (A-NAFEMS-LE10), first lines:

```
Use the supplied original STEP/Gmsh geometry definition; generate and refine your own analysis mesh.

The problem, material data, loads and constraints are supplied in published-problem.png, published-physics.txt
and input.json. The text declarations are original source excerpts, not a solution program. In those declarations
u/v/w denote x/y/z displacement. Pressure p is inward normal pressure; tension is outward normal traction. Units
and the requested sampling point are in input.json.

Solve this physical problem using FEM. Save executable main.py, mesh.msh, field output (VTK/XDMF/NPZ), solver.log
and refinement.json with at least two distinct mesh resolutions and the requested quantity at each. For cad_fem,
export part.step. Include an explanation of formulation, physical setup and stress recovery. Do not return a
recalled stress value without a solver workflow.

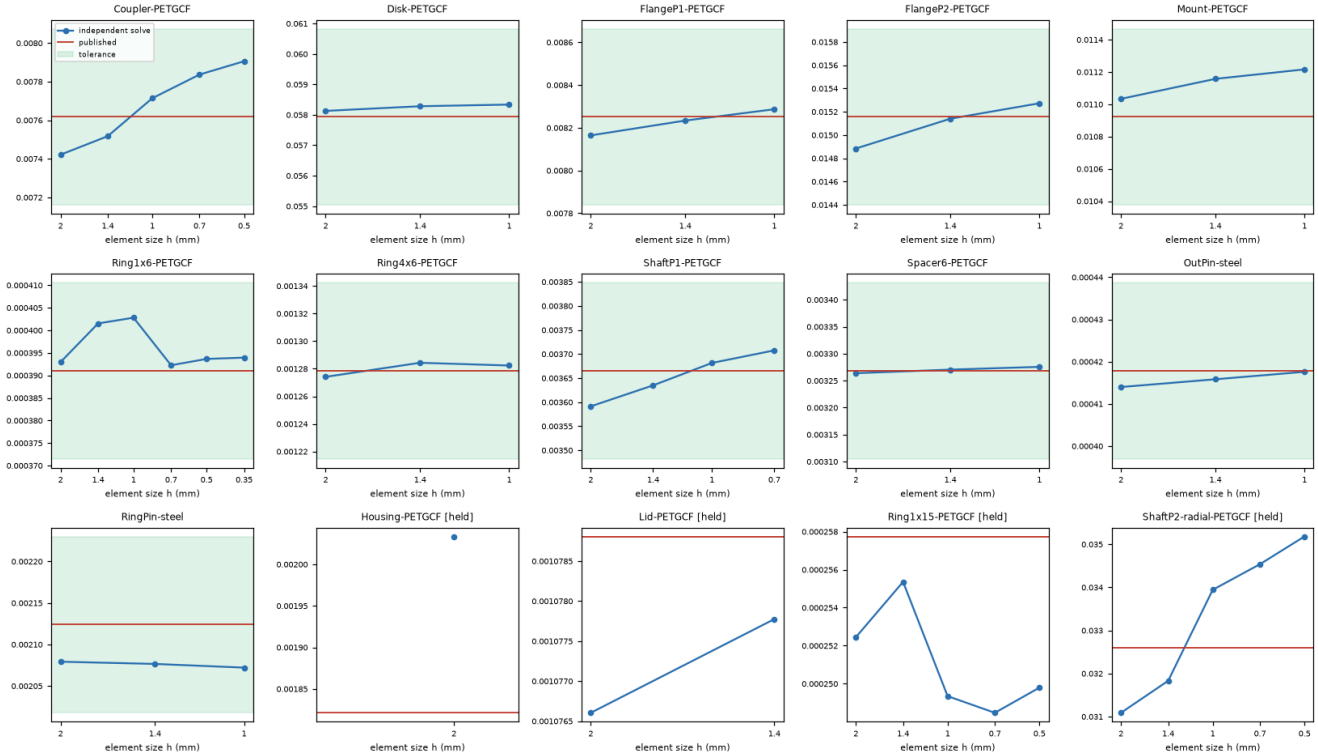
Requested quantity: signed normal stress 'sigma_yy_at_D' at the point [2000, 0, 300] (mm), reported in MPa.
input.json gives units, the sampling point and source precision notes.

Outputs:
- sigma_yy_at_D [MPa]: signed normal stress at the requested point (signed)

Submi ...
```

Flags: public NAFEMS benchmark: answer may be recalled; calibration only; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

Cycloidal convergence study (v1.2)



part	published (mm)	finest independent (mm)	difference	meshes	tolerance
Coupler-PETGCF	0.00761747	0.00790579	+3.8 %	5	6 %
Disk-PETGCF	0.0579329	0.058337	+0.7 %	3	5 %
FlangeP1-PETGCF	0.00825116	0.00828686	+0.4 %	3	5 %
FlangeP2-PETGCF	0.0151561	0.015273	+0.8 %	3	5 %
Mount-PETGCF	0.010922	0.0112161	+2.7 %	3	5 %
Ring1x6-PETGCF	0.000391032	0.000393946	+0.7 %	6	5 %
Ring4x6-PETGCF	0.00127854	0.00128233	+0.3 %	3	5 %
ShaftP1-PETGCF	0.00366527	0.0037076	+1.2 %	4	5 %
Spacer6-PETGCF	0.00326839	0.00327558	+0.2 %	3	5 %
OutPin-steel	0.00041785	0.000417598	-0.1 %	3	5 %
RingPin-steel	0.00212389	0.00207211	-2.4 %	3	5 %

Independent solves: supplied STEP remeshed with gmsh, second-order geometry, quadratic displacement, FEniCSx/PETSc MUMPS, loads as constant traction normalised to the meshed area. Admitted parts: refinement settled (last change < 1 %, previous < 2 %), force-balance residual < 1e-6 (the study script's 1e-8 cut-off is relaxed because 1e-8 to 1e-7 residuals are exact at these tolerances), band <= 10 %. Tolerance = |finest - published| / published + 2 x last change, rounded up to 1 %, at least 5 %. Held: Housing and Lid (finer meshes exceed the study's 220k-element solver guard), Ring1x15 (residual 4.7e-4, values do not settle), ShaftP2-radial (still changing 1.8 % per refinement, 7.9 % from the published value). The study files are in the evaluator evidence bundle; they never replace the published targets.

Held tasks and source registry

held tasks	reason
40	External design evaluator and environment not packaged; locally defined design variants excluded from published-source evaluation.
8	Reconstruction geometry validator and linked workflow verification not validated; retained as unscored controls/specifications.
2	Source STEP imports but fails OpenCASCADE BRepCheck validity; no silent repair.
1	Mesh-convergence study: only 1 mesh(es) solved; finer meshes exceed the study's 220k-element direct-solver guard, no converged estimate.
1	Mesh-convergence study: only 2 mesh(es) solved; finer meshes exceed the study's 220k-element direct-solver guard, no converged estimate.
1	Mesh-convergence study: force-balance residual 4.7e-04 and refinement values 0.00025245, 0.00025536, 0.00024935, 0.00024847, 0.00024979 mm do not settle cleanly.
1	Mesh-convergence study: not converged: last refinement change 1.83 %, previous 1.72 %; finest independent value 7.9 % from the published value (empirical band 12 %).

Source registry: 91 collected collections (74 held, 9 pilot, 8 excluded). Largest held pools that could become tasks with further work: Code_Aster validation cases (50 cases, 761 published checks), SFEM (61), GeoPINONet lugs (80), Bath auxetics (18).

collection	status	tasks / candidates	blocker / decision
SimJEB GE jet engine bracket (381 human designs)	pilot	36	Analysis displacement vs SimJEB published OptiStruct; peak stress/verdict and design tasks via the validated benchmark FEM grader. 13 redesign/lighten tasks without a passing human witness design (unproven feasibility) are left out.
AgentFEM bracket elasticity (176 pairs)	pilot	32	Publisher test split (32) admitted with the source STEP unit error fixed (metre-valued coordinates under a mm header; shipped copies scaled to true mm and checked against the FEM nodes); train/validation (144) kept out to avoid redundancy within one parametric family.
GeoPINONet holed plates (80 pairs)	pilot	16	Validation split (16) admitted with BCs stated by geometry (archive fixed/load STL names are swapped relative to the fields).
Morphing-wing cantilever plates (modal)	pilot	8	Tests 1-3, 5-9 admitted; Test 4 excluded (STEP fef50 % vs result block f75 %).
3D-printable cycloidal drive component FEM (31 saved solutions)	pilot	15	15 admitted (PETG-CF set + steel pins), displacement only: the source's CalculiX error estimate at the stress peaks is 11-49 %, so stress is not graded. PLA reruns of the same geometry/loads, the axial-cam variant, the mesh-only housing FRD and the uncorrected disk deck are not separate tasks.
CADWorld FreeCAD/CalculiX FEM tasks (3)	pilot	1	001 admitted; 002 left out (support on a single edge, load on an auxiliary Box object; 76.6 GPa peak is a singular-support artefact); 003 left out (20 MN on a 30 mm specimen gives 811 mm displacement, outside small-displacement analysis).
NAFEMS LE1/LE10/LE11 via FeenoX	pilot	3	Calibration tier (public benchmarks, answers may be recalled); GNL5 large-deflection case not packaged.
FeenoX power-plant valve transient thermal	pilot	1	Admitted; Gmsh surface tags verified equal to OpenCASCADE face indices.
OpenRiggingPlate 7075-T6 (7 ANSYS load tests + drawing)	pilot	6	Silver: references are rounded legend maxima; load scopes come from the source figures. Tests 4 (README/folder/figure load labels disagree) and 5 (results labelled TEST-5R, a revised set-up) left out.
COMSOL legacy studies (turbine stator, gears, muffler, ...)	excluded	7	needs a proprietary solver/material model to define the problem - COMSOL-only solved models; quantitative fields require COMSOL export.
Ansys ACP workshops (4)	excluded	4	no FEM result - No saved results.
Piston assembly (duplicate source)	excluded	0	duplicate, superseded or locally generated - Byte-identical duplicate of the piston source.
6-axis robot arm	excluded	1	no FEM result - Rigid-body joint motion only; no deformable FEM.
CAD/drawing leads without FEM output	excluded	7	no FEM result - No saved or published FEM result.
Test-bench strip restarts	excluded	4	duplicate, superseded or locally generated - Duplicate frame CAD; strip results have no matching CAD.
SFEM method re-run on ABC CAD	excluded	489	duplicate, superseded or locally generated - Loads generated locally; the user asked for real engineering set-ups, not invented loads.
Engineering FEM v1	excluded	14	duplicate, superseded or locally generated - Superseded by later collections.
DeepWheel free-free wheel modes (10 downloaded of 904)	held	10	conflicting or unmatched source references - CAD mass at the publisher density (2680 kg/m^3) is 3.25 % below the publisher CSV mass; the density behind the published frequencies cannot be determined.

collection	status	tasks / candidates	blocker / decision
FEMCAD-Hard v1 (Wolfram FEM docs)	held	0	No active tasks after v1.1 validation gates - Five tasks with closed-form/page-stated references admitted.
SFEM stochastic linear elasticity (61)	held	61	no deterministic, mesh-independent numerical reference - Fields are medians over random, unseeded load clumps whose locations were not stored: no deterministic set-up to give a model.
Bath curved-link auxetic lattices (18)	held	18	CAD only in a proprietary native format - CAD is Autodesk Inventor .ipt only; the ANSYS .dat decks carry the mesh, so no open CAD exists for a CAD->FEM task.
Auxetic meta-shells torsional buckling (6)	held	6	supports/loads/materials not recoverable as CAD-face set-up - References are torque-twist curves of nonlinear Abaqus shell models; the shell section, contact and loading must be transcribed from the decks first.
Scan-based cold-formed steel columns (3)	held	3	no B-rep CAD (mesh, scan or paper geometry only) - Geometry is a scan STL only (no B-rep); imperfections are encoded in the mesh.
NAFEMS GNL5 large-deflection beam	held	1	supports/loads/materials not recoverable as CAD-face set-up - Not in the pilot packaging; geometric-nonlinear prompt and load history not yet transcribed.
Code_Aster validation cases (50, 761 published checks)	held	50	no B-rep CAD (mesh, scan or paper geometry only) - No CAD (GIBI/MED meshes; 28 geometry scripts); checks are keyed to mesh groups. Prompts must be transcribed from the French V-documents.
FEMCAD Source v2 (Wolfram, 100 linked cases)	held	100	no deterministic, mesh-independent numerical reference - Task specifications only: no gold answers were extracted (no solver runs by policy).
COMSOL application-gallery joint tasks (50)	held	50	needs a proprietary solver/material model to define the problem - No numerical references or tolerances; source models need COMSOL.
DeepJEB brackets (2,138 scalar results)	held	2138	no B-rep CAD (mesh, scan or paper geometry only) - Only one STEP downloaded (shell, not solid); same GE load set as SimJEB, which is admitted.
FrontISTR FreeCAD benchmarks: gear, can, bike frame, spring	held	4	no deterministic, mesh-independent numerical reference - Only peak von Mises is published and it keeps rising with refinement (gear 334 -> 397 -> 442 MPa): no mesh-independent target.
GeoPINONet 3-D lugs (80)	held	80	supports/loads/materials not recoverable as CAD-face set-up - Pin-bore load distribution (uniform force vs bearing) is not documented, and the archive's fixed/load STL names are swapped.
UCSM sheet-metal inverse forming (8)	held	8	supports/loads/materials not recoverable as CAD-face set-up - LS-DYNA inverse one-step strains on surface CAD; material card and process input not supplied.
Tricuspid heart valve (Abaqus/Explicit VUMAT)	held	1	needs a proprietary solver/material model to define the problem - Shell-surface IGES and a user material subroutine; replay needs the published VUMAT inside Abaqus.
fcFEM collapse examples (5)	held	5	conflicting or unmatched source references - Saved snapshots cannot be matched to load factors/increments from the published parameter files.
Aerospace wing box (IGES + APDL)	held	1	conflicting or unmatched source references - IGES faces only; geometry/unit/set-up correspondence with the published results unresolved.
Cemented hip implant contact (Leeds)	held	1	supports/loads/materials not recoverable as CAD-face set-up - Pelvis/cement are open IGES surfaces; only one of four wear states maps to an executable deck.
HIRENASD wing modes (NASA)	held	1	no deterministic, mesh-independent numerical reference - Mode-shape fields only; frequency file not public; frame correspondence unresolved.
Blade dovetail reduced model (PoliTo)	held	1	no deterministic, mesh-independent numerical reference - Reduced mass/stiffness matrices only, no response results.
Fokker F28 fuselage crash (TU Delft)	held	3	no deterministic, mesh-independent numerical reference - Results only as plots mixing FEM and experiment; no machine-readable response.
WAAM wall thermal history and distortion (AIMEN/SCAI)	held	1	supports/loads/materials not recoverable as CAD-face set-up - No solver deck (moving heat source parameters only in PDFs); VMAP unit tags undefined.
Capriccio FE/MD fracture, Digimat cure, bimetal strip, SOFA actuator, tapered spring	held	6	needs a proprietary solver/material model to define the problem - Coupled FE/MD or proprietary-material (Digimat/Abaqus) set-ups; no open CAD; fields partly non-finite.
FreeCAD Code_Aster laminate + CalculiX comparison; Hannover milling	held	3	conflicting or unmatched source references - Laminate result units and touched solver objects unresolved; milling has measurements, not FEM.
Test-bench frame (FreeCAD/CalculiX)	held	1	no deterministic, mesh-independent numerical reference - Displacement is prescribed; the only response target is a coarse-mesh peak stress.

collection	status	tasks / candidates	blocker / decision
ACP composite kiteboard (Ansys tutorial)	held	1	needs a proprietary solver/material model to define the problem - Layup lives in ACP; 'SOLVED' archive lacks results; reference from a different software version.
PrePoMax tutorial geometries (12)	held	12	no deterministic, mesh-independent numerical reference - Idealised tutorial parts with plot-only results in a manual.
Modular helical marine anchor (OpenEI, 3 load sets)	held	3	CAD only in a proprietary native format - CAD only as DesignModeler/SolidWorks/Parasolid; analysis uses idealised equivalent members.
PyDPF composites examples (5)	held	5	CAD only in a proprietary native format - Native SpaceClaim geometry, ACP layups; parser failures on two projects.
Layered composite pipe (2 revisions)	held	1	conflicting or unmatched source references - Two conflicting saved revisions; orthotropic shell layup; IGES is a surface.
Pneumatic multi-link gripper (nonlinear contact)	held	3	supports/loads/materials not recoverable as CAD-face set-up - Joint/contact definitions not recoverable without the 2026R1 archive decoder; baseline/A1 have tables only.
Fin Ray 3-finger hyperelastic contact	held	1	no B-rep CAD (mesh, scan or paper geometry only) - Full three-finger CAD not published (one finger STEP + assembled mesh).
Bolted nozzle-flange sector (3 repos)	held	4	CAD only in a proprietary native format - DesignModeler-only CAD; README vs deck conflicts (units, materials, temperatures).
Formula SAE chassis (4 static cases)	held	4	CAD only in a proprietary native format - SpaceClaim-only beam CAD; decoded fields conflict with the published displacements.
Go-kart chassis modal/harmonic	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; plot/report references.
IIT Mandi shear frame (4 modal variants)	held	4	CAD only in a proprietary native format - Abaqus CAE-only CAD; joint modelling and damage-case revision mapping unresolved.
Double-wishbone suspension	held	1	conflicting or unmatched source references - Supplied CAD revision cannot be matched to the baseline or redesigned stress state.
Perforated brake disc steady thermal	held	1	supports/loads/materials not recoverable as CAD-face set-up - Peak is a prescribed boundary value; convection/temperature face scopes must be mapped from the deck.
Horizontal pressure vessel (Nimonic 75)	held	1	supports/loads/materials not recoverable as CAD-face set-up - IGES surfaces only; saddle support scope undocumented.
Pneumatic cylinder cover	held	1	supports/loads/materials not recoverable as CAD-face set-up - Source support is a single fixed node plus frictionless node sets; not expressible as CAD-face BCs.
IC engine connecting rod (2 saved cases)	held	2	supports/loads/materials not recoverable as CAD-face set-up - Traction/support scopes must be mapped from the ANSYS deck node components to STEP faces.
Connecting rod (SolidWorks/ANSYS report)	held	1	conflicting or unmatched source references - STEP in inches with a 4.98 m extent; plots in CGS; support scope undocumented.
Twin-wheel nose landing gear	held	1	supports/loads/materials not recoverable as CAD-face set-up - Only a minimum safety factor is published; load/support scopes only in report figures; materials partly unconfirmed.
Single-cylinder crank assembly	held	1	supports/loads/materials not recoverable as CAD-face set-up - Material cards undocumented; thermal references conflict.
Ventilated brake rotor (thermal + structural)	held	1	supports/loads/materials not recoverable as CAD-face set-up - IGES faces with no solid; ten fixed faces and pressure scopes not identified.
Pipe-flange assembly (2 connection models)	held	2	CAD only in a proprietary native format - SolidWorks-only CAD; load scope undocumented; drawing inch/mm conflict.
Bolted shelf with pretension and contact	held	1	conflicting or unmatched source references - SpaceClaim-only CAD; saved results disagree with the report (0.028 vs 0.025 mm).
CubeSat frame archive	held	1	conflicting or unmatched source references - Archive geometry does not match the published CubeSat.
Elmer conjugate cooling (radiator, tube)	held	2	conflicting or unmatched source references - Set-up lives in native FreeCAD/Elmer objects; README vs native heat inputs differ; fields undershoot ambient.
Titanium wing static + prestressed modal	held	2	CAD only in a proprietary native format - DesignModeler-only CAD.
Wheel rim static + modal (5 analyses)	held	5	CAD only in a proprietary native format - SolidWorks/SpaceClaim-only CAD.
Quadcopter frame static/modal/harmonic	held	3	CAD only in a proprietary native format - Native assemblies only; harmonic load cards conflict with the report; modal uses point masses.
Curved steam pipe and elbow thermal	held	4	CAD only in a proprietary native format - SpaceClaim-only CAD; problem figures not complete drawings.

collection	status	tasks / candidates	blocker / decision
Seven-fin heat sink steady thermal	held	1	supports/loads/materials not recoverable as CAD-face set-up - Convection scope (41 of 55 faces) not identifiable; the published temperatures imply ~0.047 m^2 of convective area, less than the fin sides alone (0.054 m^2).
NX fin-array heat sink	held	1	CAD only in a proprietary native format - NX-only CAD; heat input and film coefficients absent.
Spur gear pair	held	1	supports/loads/materials not recoverable as CAD-face set-up - Load, material and contact law not specified.
Spring shock absorber	held	1	supports/loads/materials not recoverable as CAD-face set-up - Loads and supports not documented.
Screw jack	held	1	conflicting or unmatched source references - README and plot references conflict; native CAD only.
Knuckle joint	held	1	CAD only in a proprietary native format - Inventor-only CAD; STEP not uploaded.
Multi-leaf spring modal	held	1	conflicting or unmatched source references - Two conflicting frequency sets; surface-only IGES.
Steel dogbone static/buckling control	held	2	CAD only in a proprietary native format - Discovery-only CAD; supports undocumented.
Piston-pin-connecting rod	held	1	conflicting or unmatched source references - Part files missing from the assembly; README conflicts.
Piston thermal-barrier coating	held	3	conflicting or unmatched source references - Export/report revision mapping unresolved; obsolete result state.
Heat-assisted incremental forming	held	6	conflicting or unmatched source references - Partial CAM CAD only; conflicting job tables.
Kresling origami	held	1	CAD only in a proprietary native format - Inventor/Parasolid-only CAD; curve-to-case mapping unresolved.
Bolted plywood truss	held	1	supports/loads/materials not recoverable as CAD-face set-up - Idealised 2-D truss export with unknown units.
Haunched steel portal frame	held	4	CAD only in a proprietary native format - Native CAD only; inch/mm drawing conflict; load scope ambiguous.
Aquantis tidal turbine	held	1	conflicting or unmatched source references - CAD/FEM pairing not established (45 m CAD vs 41.5 m FEM).
Splined differential shaft	held	1	conflicting or unmatched source references - Report displacement exceeds the part length ~963x; conflicting reports.
Carbon-fibre UAV frame (62 bodies)	held	1	supports/loads/materials not recoverable as CAD-face set-up - Orthotropic properties and 132 bonded contacts not frozen; report-only references.
External gear-pump housing	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; configuration/plot pairing unknown.
Ribbed mounting flange (drawing + 55-row table)	held	1	conflicting or unmatched source references - Drawing dimension conflicts; screenshot values match no table row; supports undocumented.
Paper drawings with FEM (pump shaft, ...)	held	10	no B-rep CAD (mesh, scan or paper geometry only) - Paper-only geometry and results; solver set-ups incomplete or inconsistent.
TU Wien e-bike frame	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; plot-only, incomplete solver specification.

Files: candidate/ (give to the model), evaluator/ (keep hidden: references, registry, scorer, review protocol, evidence bundle), held/ (held tasks with their inputs and the excluded local targets), report/, scripts/. Licences differ per source and several sources are local-research-only: do not publish the package as a whole. No model results yet.